F79MB Statistical Models B

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1. Review of estimation and testing

Example 1.1:
The reaction time in athletics racing is measured as the time taken from the starting signal of the race to the moment an athlete has left the starting blocks. In a juniors championship, the reaction times of forty athletes racing in the 800 metres event were recorded and are given below (in seconds):

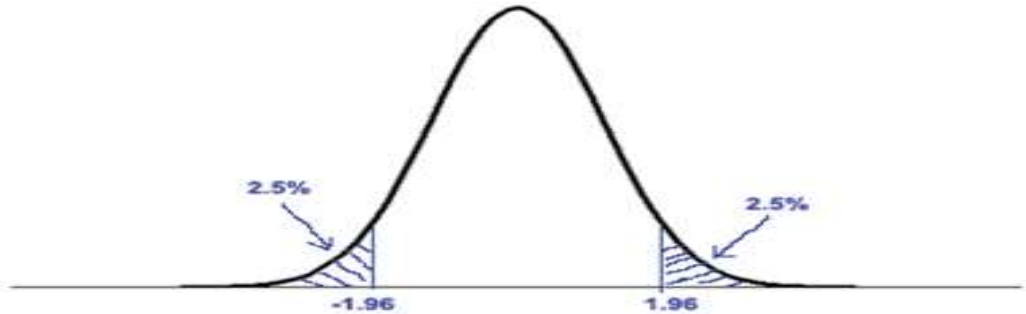
0.341	0.368	0.358	0.335	0.347	0.373	0.365	0.332
0.330	0.362	0.364	0.352	0.338	0.327	0.350	0.358
0.385	0.370	0.350	0.341	0.349	0.354	0.330	0.352
0.377	0.355	0.324	0.375	0.338	0.335	0.382	0.355
0.338	0.373	0.352	0.385	0.349	0.359	0.378	0.346

Determine a 95% confidence interval for the mean reaction time for this event.

Sample mean = 0.3538 Sample standard deviation = 0.01685

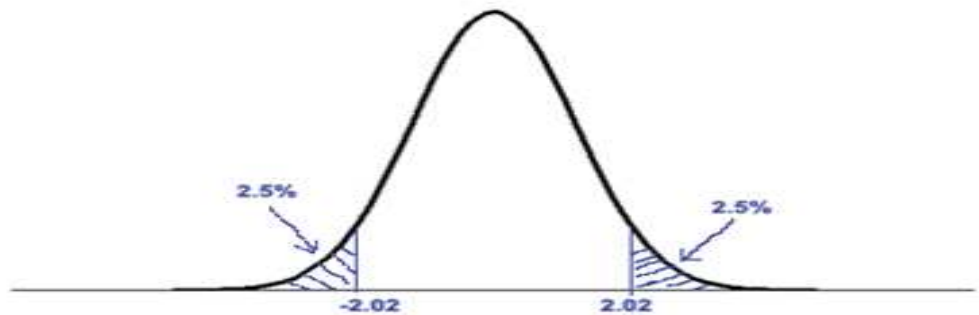
Do we know the distribution of the data?

(a) Since the sample size is 40, we can use critical values from the standard normal distribution employing the *Central Limit Theorem*:



Find that 95% CI for μ is (0.3486, 0.3590).

(b) If the sample comes from a normal distribution, we can also use critical values from the t_{39} distribution



Find that 95% CI for μ is now (0.3484, 0.3592).

Note: t-distribution CI slightly wider than normal distribution CI, because t-distribution takes into account extra uncertainty from estimating σ.

Two ways that we can check whether or not the sample is normal are stem and leaf plots and QQ plots. (QQ plots covered in chapter 2)

Example 1.2:
A particular stretch of motorway (in a European country), has speed limit 100 kilometres per hour (kph). The speed of 20 vehicles travelling on this part of the motorway was recorded and gave the following measurements (in kph):

```
115  82   96   87  100
 69 108  116  180   99
 94  86  104   98  103
102  62   94  126   96
```

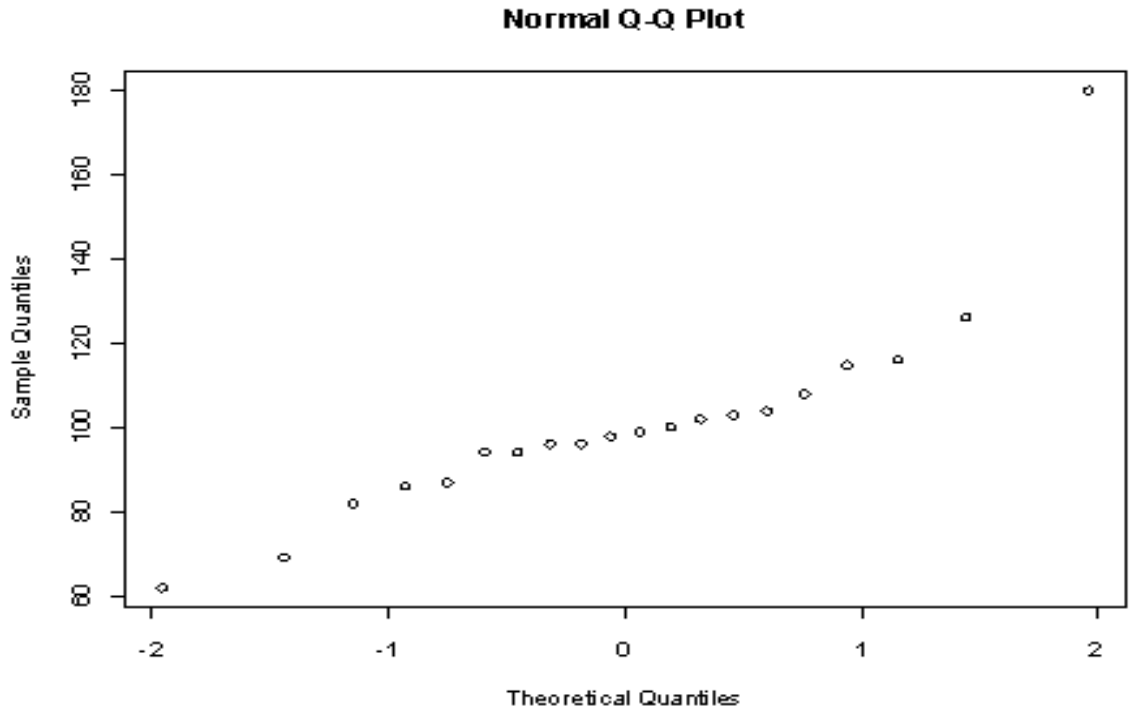
Calculate 95% confidence intervals for the true mean and s.d. of the speed of vehicles travelling on this part of the motorway.

Sample mean = 100.85 Sample standard deviation = 23.8841

Answer:

Are the data normally distributed?

QQ plots will be covered in chapter 2. For now, just need to know that for normal-distributed data, the normal QQ plot will be roughly a straight line.



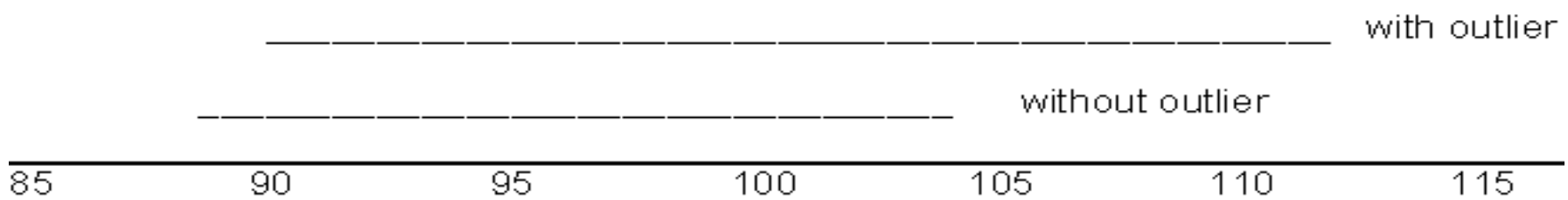
- The QQ plot shows that normality is plausible except for an outlier at 180.
- We will assume that it has been wrongly recorded and should, in fact, be 80. [Be very cautious of making this kind of assumption!]
- Calculations will be done twice, once with original value then with "corrected" value.

Using *original data*:
95% CI for mean is (89.67, 112.03)
95% CI for sd is (18.16, 34.88)

Now we will repeat the process using the *"corrected" value*:

Sample mean = 95.85 Sample standard deviation = 15.40

95% CI for mean (88.6,103.1)
95% CI for sd (11.71, 22.50)



95% CIs for mean

The outlier has caused increases in both location and spread.

A CI for the median

Resistant measure: One that is not influenced by extremely high or extremely low data values.

The median tends to be more resistant than mean, and interquartile range tends to be more resistant than standard deviation.

Example 1.3:
Calculate a 95% confidence interval for the **median** of the results given in the 'eurodist' dataset in R. The data give the road distances (in km) between 21 cities in Europe. The data are taken from a table in The Cambridge Encyclopaedia.

```
>data(eurodist)
```

Part of the table of distances is shown below.

	Ath	Bar	Bru	Cal	Cher	Col
Barcelona	3313					
Brussels	2963	1318				
Calais	3175	1326	204			
Cherbourg	3339	1294	583	460		
Cologne	2762	1498	206	409	785	
Copenhagen	3276	2218	966	1136	1545	760

General theory:

A possible estimator of the population median (m) is the sample median $\hat{m}$.

Asymptotically, i.e. for a large sample, it can be shown (we won't prove this) that approximately

$$\hat{m} \sim N\left(m,\ \left(\frac{1}{2f(m)\sqrt{n}}\right)^2\right)$$

where $f(m)$ is the density of the underlying population at the median m. For computation purposes $f(m)$ may need to be estimated, for instance by $f(\hat{m})$.

The standard error of $\hat{m}$ is, from above, $s.e.(\hat{m}) = \frac{1}{2f(m)\sqrt{n}}$

For a *normally distributed population*, the standard error of $\hat{m}$ simplifies to $s.e.(\hat{m}) = \sigma\sqrt{\frac{\pi}{2n}}$

Proof: see video.

(Or see the extra file for Example 1.3 given on Canvas).

Note that s.e.(sample median) $= \sigma\sqrt{\frac{\pi}{2n}} = \sqrt{\frac{\pi}{2}}\frac{\sigma}{\sqrt{n}} \approx 1.25\frac{\sigma}{\sqrt{n}}$
and recall that s.e.(sample mean) $= \frac{\sigma}{\sqrt{n}}$

So, in the case of (approximately) normally distributed data, the median is a less efficient estimator than the sample mean (confidence intervals are wider).

However, for longer-tailed distributions the sample median is a more efficient estimator of location than the sample mean.

Example 1.3 answer:

Pairwise distances between 21 cities, so sample size $n = \frac{21\times 20}{2} = 210$

Sample mean = 1505.148 Median = 1311.5 Standard deviation = 898.7842

Approximate normality seems reasonably plausible, although with some evidence of skewness and outliers (check with stem and leaf plot).

So the 95% confidence interval for the median is (1159.1, 1463.9)

For comparison, a 95% CI for the mean is $(1383.6,\ 1626.7)$

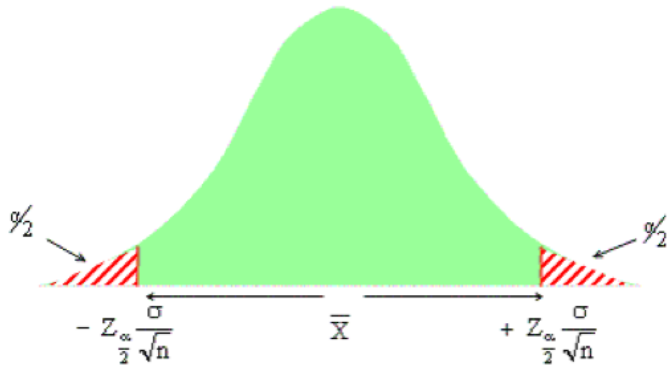
Note that CI for mean is narrower than CI for median.

Chapter 1 recap

For Mean with n large ($n \geq 30$, say)	Use Z-distribution Confidence Interval	C.I. $= \bar{X} \pm Z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$
For Mean with unknown variance, Normal data, and n small ($n < 30$, say)	Use t-distribution Confidence Interval	C.I. $= \bar{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$
For Variance	Use χ^2-distribution Confidence Interval	C.I. $= \left[\frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}} \right]$
For Proportion if $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$	Use Z-distribution Confidence Interval	C.I. $= \hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Summary

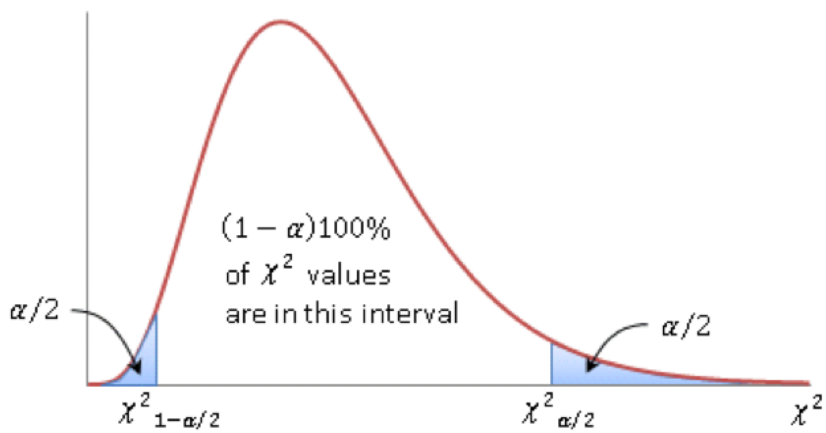
$100(1-\alpha)\%$ CI for μ:



$$\text{C.I.} = \bar{X} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

where $\bar{X}$ = the sample mean
σ = the population standard deviation
$Z_{\frac{\alpha}{2}}$ = the Z value for the desired confidence level α (obtained from an Area Under the Normal Curve table)

$100(1-\alpha)\%$ CI for σ^2:



Median:

Definition of the population median (m)

If a continuous random variable $X \in (-\infty, \infty)$ follows a distribution with probability density function $f(x)$, then the median, m, is defined as the value such that

$$\int_{-\infty}^{m} f(x)\, dx \quad = \quad \frac{1}{2}$$

(i.e. it is the value that splits the probability distribution into 2 equal halves).